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On the utilization controversy in the demand-led growth literature: A quantile unit root approach☆

Guilherme de Oliveira *

Federal University of Santa Catarina, Brazil

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ABSTRACT

The demand-led growth literature is embroiled in a controversy surrounding the long-run stationarity of capacity utilization. One perspective posits that a non-stationary rate of capacity utilization implies its endogeneity to aggregate demand, aligning with the neo-Kaleckian model. In contrast, an opposing viewpoint suggests the possibility of long-run stability independent of demand fluctuations. Conventional empirical approaches generally support the latter viewpoint. This study proposes a novel way to estimate capacity utilization persistence by using a quantile autoregression (QAR) model, which allows for the possibility that shocks of different signs and magnitudes impact capacity utilization differently. Using official survey measures from developed and developing countries, we find evidence in favor of local unit root tendencies, with recessions exhibiting persistent effects on utilization. Additionally, we identify an asymmetric capacity utilization response, where, in general, recessive shocks are persistent while positive shocks are transitory. These results provide a fresh perspective on how capacity utilization responds to different economic conditions.

1. Introduction

The neo-Kaleckian model is considered one of the most influential contributions to the post-Keynesian theory of economic growth and income distribution. Due to its elegant and tractable mathematical formulation, the model has undergone numerous extensions since the original contributions by Rowthorn (1981), Dutt (1984), and Taylor (1985). At the same time, however, the model has not been free from severe theoretical and empirical criticism over the years.

A central critique is that in the long run the *actual* rate of capacity utilization does not converge to its *desired* or *normal* rate. In the model, the rate of capacity utilization is an accommodating variable (Skott, 2010). The argument is that the actual utilization cannot be persistently different from its desired rate. One reason is that the maintenance of capital (under)overutilization during a long period is unprofitable, thus being inconsistent with an optimal response of the firms. The other refers to Harrodian instability: unless the actual and desired rates are equal in the long run, firms will not be in steady-state equilibrium (Auerbach and Skott, 1988). As there is nothing in the neo-Kaleckian model to bring the actual rate towards its desired rate, its predictions would apply only to the short and medium run (Nikiforos, 2016).

A response was articulated by Amadeo (1986), Lavoie (1995, 1996), Dutt (1997), and Lavoie et al. (2004). In essence, the argument conceded the equalization between the actual and desired rate of capacity utilization in the long run. However, it is the desired rate that is endogenous to the actual rate. The response is logically consistent, but at least two critiques remain. The model fails to explain why any deviation of the actual utilization from the desired utilization will induce firms to revise their desired rate (Skott, 2012), and shocks to demand still have persistent effects on actual utilization. In other words, the model predicts a non-stationary rate of capacity utilization. The empirical evidence does not support this prediction (Schoder, 2014).

In the case of the United States, for example, the conventional empirical approach usually uses the Federal Reserve data on capacity utilization. The pattern of these series, as interpreted by the literature, reveals that the rate of capacity utilization fluctuates around a constant mean (the actual long-run level), approximately 80% (Nikiforos, 2016). This pattern is observed, *mutatis mutandis*, in other countries. The argument is then that despite exhibiting significant short-run variations, the

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* Correspondence to: Department of Economics and International Relations, Federal University of Santa Catarina, R. Eng. Agronômico Andrei Cristian Ferreira, s/n. Trindade, Florianópolis SC, 88040-900, Brazil.

E-mail address: oliveira.guilherme@ufsc.br.

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actual utilization does not deviate persistently from the long-run level. Conventional unit root tests usually support this claim by verifying the stationarity of these series. Gahn and González (2022), for instance, show that the direct survey measures of capacity utilization of 24 developed and developing countries are stationary (not persistent), positively correlated with growth in the short run, but uncorrelated with growth in the long run.

The literature dealing with the presence of unit roots in the rate of capacity utilization focuses on constant-coefficient time series models that rely on a single measure of conditional central tendency, the mean. Consequently, structural and demand-led changes are homogeneously treated in the conditional distribution of capacity utilization. The utilization realizations that are high or low relative to utilization realizations in the previous periods are not distinguished. But, it is reasonable to suspect that the capacity utilization response to a large negative shock might be qualitatively different from a median or a large positive shock. In other words, firms may asymmetrically adjust the utilization rate depending on the intensity and sign of the shock.

If the asymmetric dynamics characterize the behavior of firms, one may investigate whether the evidence of persistence in capacity utilization fluctuations also varies according to the intensity of the shock. The idea of qualitative different dynamics along the business cycles dates back at least to Mitchell (1927) and Keynes (1936). In the General Theory, for instance, Keynes suggests that the length of time between a peak and a recession is much shorter than the length of time from a recession to recovery. A large empirical literature has taken this possibility into account in the case of GDP fluctuations. Some studies find evidence of the permanent effects of recessions and transitory effects of expansions (Hamilton, 1989; Hosseinkouchack and Wolters, 2013), while others suggest the opposite (Beaudry and Koop, 1993). The present study contributes to the literature on demand-led growth by examining this nonlinear behavior in the context of capacity utilization fluctuations.

This study proposes a novel way to analyze the evidence of persistence in capacity utilization fluctuations. An empirical exercise with official survey measures of capacity utilization from 15 developed and developing countries illustrates the method. The empirical strategy tests the presence of a unit root at the tails of the conditional distribution of capacity utilization by using a quantile autoregression (QAR) model, which offers advantages over conventional approaches used by Schoder (2014), Nikiforos (2016) and Gahn and González (2022). Similar approaches have been successfully applied in various economic contexts, including examining the persistence of inflation (Valera et al., 2017; Gaglianone et al., 2018). The QAR model allows for the study of asymmetric dynamics and local persistence in time series, as emphasized by Koenker and Xiao (2004). It is important to note that even when specific occurrences of mean reversion are sufficient to ensure global stationarity, the empirical evidence regarding the utilization controversy suggests that certain forms of the model demonstrate that time series can exhibit unit root tendencies or even temporary explosive behavior (Koenker and Xiao, 2006). Therefore, our approach represents a significant advancement over conventional methods used in the literature, enabling a deeper understanding of the nuanced dynamics of capacity utilization.

The empirical model considers that the response of the rate of capacity utilization to exogenous shocks is contingent upon the phase of the business cycle. To capture the inherent asymmetry in these dynamics, we explicitly account for the differential impact of shocks with varying signs and magnitudes on capacity utilization. Our analysis uncovers significant asymmetry in capacity utilization fluctuations. Specifically, shocks located below the median, which are associated with recessionary periods, are found to be persistent. In contrast, shocks located at the median and upper conditional quantiles are transitory. These new results have significant implications and introduce a new dimension for the ongoing utilization controversy discussed in previous studies by Setterfield and David Avritzer (2020), Gahn and González (2020) and Nikiforos (2020). While strong negative shocks suggest that capacity utilization behaves as predicted by the neo-Kaleckian model in terms of its long-run adjustment, positive variations above the median suggest that model fails to explain the observed behavior of capacity utilization. These findings emphasize the necessity for a comprehensive understanding of the factors that influence capacity utilization dynamics in diverse economic contexts.

The remainder of the paper is organized as follows. Section 2 presents a Neo-Kaleckian baseline model. Section 3 details the identification strategy. Section 4 discusses the empirical results. The paper closes with a summary of the main conclusions derived along the way.

2. A Neo-Kaleckian benchmark model

This section presents a Neo-Kaleckian benchmark model to briefly summarize some of the main controversies and highlight key empirical predictions surrounding the adjustment mechanism in the long run. It is beyond the scope of this study to present a detailed description of the model and its empirical regularities (see, e.g., Hein et al. (2012) and Gahn and González (2022)).

A substantial literature discusses the failure of the neo-Kaleckian model to equate the actual rate of capacity utilization with the desired rate. Kurz (1986), Committeri (1986), Duménil and Lévy (1999), and Auerbach and Skott (1988) are some of the early critics. The critique is well-known and can be summarized as follows. First, the maintenance of a persistent capital under(over)utilization is unprofitable in the long run, and thus inconsistent with an optimal response of firms. The desired rate of capacity utilization is determined within the cost-minimizing long-run problem of the firm. Finally, in the long run, the actual rate fluctuates around an exogenous and structurally determined desired utilization rate (Nikiforos, 2016).

Some neo-Kaleckian responses to the critique suggest that the adjustment process based on “the distinction between the actual utilization and a given desired utilization rate may be implausible” (Dutt, 1990, p. 59), while others face the problem and develop analytical models. A formal response was articulated in a series of contributions made by Amadeo (1986), Lavoie (1995, 1996), Dutt (1997), and Lavoie et al. (2004). To examine some implications of this response, a concise presentation of a neo-Kaleckian model for a closed economy without governmental activities is sufficient. The presentation is based in Lavoie (1995) and Dutt (1997).

The short-run structure of a closed economy model without monetary and fiscal activities can be described by the following set of equations:

$$g^I \equiv \frac{I}{K} = \eta + \eta_u(u - u^d) + \eta_\pi \pi, \tag{1}$$

$$g^S \equiv \frac{S}{K} = s\pi u\rho, \tag{2}$$

$$\pi = \bar{\pi} \equiv 1 - \sigma, \tag{3}$$

$$g^I = g^S, \tag{4}$$

$$g \equiv \hat{K} = g^I - \delta. \tag{5}$$

Eqs. (1) and (2) are the investment, I, and saving, S, functions, both normalized by the capital stock, K. Investment is increasing in the profit share, π, and the deviation of the actual utilization, u, from its desired rate, u^d. $\eta_u, \eta, \eta_\pi \in \mathbb{R}_+$ are exogenously given and fixed parameters. In (2), s is the savings rate, while ρ denotes the technical output-capital ratio. In (3), the markup rate is fixed and the classical conventional wage share assumption, in which the wage share, $1 - \sigma$, is exogenous, is supposed to hold. The equilibrium condition in the product market is defined by (4). Finally, (5) defines the growth rate of K as being equal to the gross accumulation minus the rate of depreciation, $\delta \in \mathbb{R}_+$. Both equations are derived under the classical definition of the profit rate, $r = \pi u \rho$.

Using (1) and (2) in (4), the equilibrium rate of capacity utilization, u, is equal to:

$$u = \frac{\eta + \eta_\pi \pi - \eta_u u^d}{s\pi\rho - \eta_u}, \tag{6}$$

in which the Keynesian stability condition, $s\pi\rho > \eta_u$, is supposed to hold; it is supposed further that $\eta + \eta_\pi \pi > \eta_u u^d$. Violation of the Keynesian stability condition can shed some light on the Harrodian instability problem (Skott, 2010). The existence of surplus labor in combination with excess capital capacity implies that u is the variable that adjusts to clear the goods market.

In turn, using (6) and (1) in (5), the equilibrium growth rate in the short run is given by:

$$g = \frac{s\pi\rho(\eta + \eta_\pi \pi - \eta_u u^d)}{s\pi\rho - \eta_u} - \delta. \tag{7}$$

Eqs. (6) and (7) define the short-run equilibrium, in which the desired utilization is supposed to be given. It is readily seen that demand-led changes, for example, changes in the autonomous investment, η, will be accommodated by changes in the actual rate of capacity utilization. In turn, changes in the actual utilization will induce the investment rate and increase the equilibrium growth rate. This adjustment is a particular feature of the model, very distinct from Robinsonian models, in which shocks are accommodated by changes in the markup rate.

In the long run, the model supposes that the short-run equilibrium values of the variables are always attained. Following Amadeo (1986), which suggests that faced with a gap between actual and desired utilization firms revise their desired rate of capacity utilization, the following dynamic equation can be formalized:

$$\frac{du^d}{dt} = \mu(u - u^d), \tag{8}$$

in which $\mu \in \mathbb{R}_+$ is a parameter that measures the speed of adjustment. Dutt (1997) and Lavoie (1995) follows a similar formalization. Dutt (1997) justifies the adaptative behavior in (8) in terms of the strategic behavior of the firms. Firms will reduce their desired rate of capacity utilization if they expect the future rate of entry to be higher than the present rate of entry. In a series of contributions Lavoie (1995, 1996) and Lavoie et al. (2004, p. 133), in turn, argues that the uncertainty surrounding the effective demand effects might "hinder firms from achieving the target unless the normal rate is itself a moving target influenced by its past values". Under this path dependence, the desired rate is thus, for firms, a "convention".

To complement the dynamical system, the Amadeo–Lavoie–Dutt argument supposes that the desired utilization changes endogenously with the expected growth rate, $\gamma = \eta + \eta_\pi \pi$. The following dynamic equation is supposed:

$$\frac{d\gamma}{dt} = \theta(g^I - \gamma), \tag{9}$$

in which $\theta \in \mathbb{R}_+$. Note that when the gross accumulation exceeds the expected rate, firms will rise the expectations about the growth rate. The opposite is true when $g^I < \gamma$. Using (1), (9) can be rewritten as benig equal to:

$$\frac{d\gamma}{dt} = \theta\eta_u(u - u^d). \tag{10}$$

Eqs. (10) and (8) form a two-dimension dynamical system, whose stationary solution is given by: $\gamma = s\pi\rho u^d$. The properties of the dynamical system can be qualitatively analyzed using conventional local stability criteria. In this sense, the determinant of the Jacobian matrix is zero; the two functions coincide. The long-run stability condition requires that: $\mu s\pi\rho > \theta\eta_u$.

For illustrative purposes, Fig. 1 presents a simulation of the trajectories of the key variables towards the long-run equilibrium. The causality runs as follows. A demand shock is accommodated by a change in the actual rate of capacity utilization in the short run, where the desired rate is supposed to be given. In the long run, it is supposed that the short-run values of the actual rate are always attained. Hence, the demand shock creates a gap between the actual and the desired utilization. The firms will revise their desired rate of capacity utilization upwards (panel (a)). The expected growth rate also grows proportionally to the utilization gap. As the gap is decreasing over time, the expected growth rate increases at a diminishing rate (panel (b)). The adjustment continues towards the steady state in which $u = u^d$.

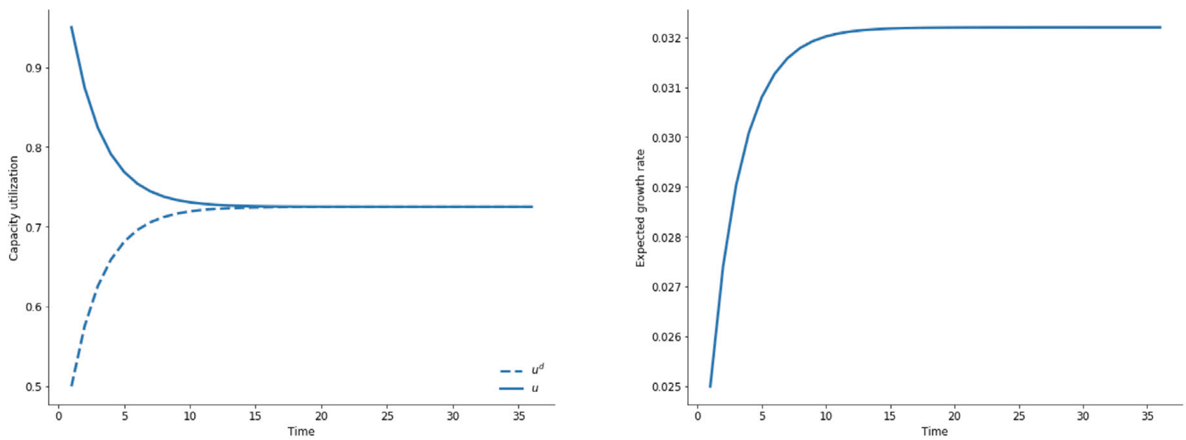
The neo-Kaleckian response is logically consistent and offers a solution to the problem of equating the actual and the desired rate of capacity utilization in the long run. However, the causality runs from the actual rate to the desired rate. Skott (2012) raises several objections to the neo-Kaleckian response, with emphasis on the lack of coherence of the behavior foundation of (8). Skott (2012, p. 120) analyzes that the argument in Lavoie (1995) and Amadeo (1986) "does not go beyond stating the formal possibility of an adjustment". The author sees the argument in Lavoie et al. (2004) as misguided, once the presence of uncertainty and conventional elements in the demand effects does not justify that the desired utilization be defined as a conventional variable. Regarding the argument in Dutt (1997) and Skott (2012) classifies it as reasonable, although the sensitivity of the desired rate to the entry threat is likely to be relatively low.

Hence, in the Amadeo–Lavoie–Dutt model, the actual utilization remains an accommodative variable in the long run. The model still predicts persistent long-run variations in u. In the canonical version of the model, this persistence represents a permanent gap between u and u^d. Now, the persistence pattern relies on the endogeneity of the desired utilization. Fluctuations in the actual rate are supposed to be able to change the long-run level of the desired rate. In addition, the modification does not result in a testable empirical specification. However, the adjustment of capacity utilization and its stationarity condition can be well described by an autoregressive process, which is detailed in the next section.

3. Methods and data

The empirical strategy is concerned with the identification of asymmetric dynamics and local persistence in capacity utilization fluctuations by using a QAR model. The first issue in this regard is related to the measure of capacity utilization. We use the official survey measures of capacity utilization. This choice is not free from criticism. Nikiforos (2016, 2021) formalizes a compelling argument that these measures, like those constructed by the U.S. Federal Reserve, are stationary by construction and closer to a cyclical indicator of economic activity than a measure of long-run variations in desired utilization. In fact, the measure problems are real, but one would need a rigorous argument to change the empirical measure of capacity utilization that is used as a central element of the criticism of the neo-Kaleckian adjustment mechanism. Gahn and González (2020), for instance, stands firm that the use of survey measures of capacity utilization is a reasonable choice, but it is the neo-Kaleckian model that fails to predict the global stationarity of these series. As in every empirical debate, the controversies on the right data will probably remain. In the present analysis, the survey measures of capacity utilization justify for comparative purposes.

The second issue is related to the time series. Given the nature of the long-run analysis, and for reasons that will become clear later, the empirical strategy demands a relatively large number of observations. The number of countries for which the data is available at a relatively long time horizon is scarce. Hence, the empirical strategy uses an availability sample of monthly and quarterly data on capacity utilization. The selection criteria define a minimum number of 150 observations. We just consider time series prior to the Covid-19 pandemic. The monthly sample is composed of Argentina, Brazil, Mexico, Colombia,



(a) Actual and desired utilization rate. (b) Expected growth rate.

Parameters: $\eta = 0.038$, $\eta_\pi = 0.05$, $\pi = 0.4$, $s = 0.2$, $\rho = 1$, $\eta_u = 0.04$, $\mu = 0.02$, $\theta = 0.01$, $u^d(0) = 0.5$, $\gamma(0) = 0.027$, $t = [0, 100]$.

Fig. 1. The behavior of the model in the long run.

Table 1
Descriptive statistics for the monthly and quarterly rate of capacity utilization.
Source: Multiple sources; See Section 3 p. 8.

Country	Time	Mean	Std. Dev.	1° Decil	9° Decil	Skew.	Kurt.	Shapiro–Wilk
Argentina	Jan2002–Jan2020	69.61	6.92	60.06	77.91	−0.43	0.12	0.9791*
Brazil	Jan1980–Jan2020	79.58	3.60	74.30	84.00	−0.40	−0.86	0.9572*
Colombia	Jan1998–Dec2019	75.98	3.77	71.73	81.19	−0.31	0.23	0.9875**
Mexico	Jan1998–Jan2020	73.07	2.16	70.36	75.77	0.36	1.66	0.9807*
Peru	Jan1994–Nov2019	65.09	18.03	42.27	87.09	0.00	−1.52	0.9087*
Philippines	Jan2002–Dec2019	81.70	2.44	78.50	84.19	−1.14	0.73	0.8595*
Thailand	Jan2000–Feb2020	63.34	4.26	58.24	68.08	−1.02	2.36	0.9448*
Turkey	Jan1987–Jan2020	75.91	3.34	71.88	79.68	−0.91	1.99	0.9521*
United States	Jan1967–Dec2019	80.10	4.24	74.98	85.54	−0.18	−0.18	0.9815*
France	Q11976–Q12020	83.60	6.25	81.10	86.40	−1.25	4.55	0.9096*
Germany	Q11960–Q12020	83.98	3.93	78.60	88.30	−0.89	0.97	0.9524*
Italy	Q41968–Q12020	75.22	3.11	71.00	78.60	−0.60	1.32	0.9728*
New Zeeland	Q21961–Q42019	89.40	2.33	85.99	92.21	−0.49	−0.49	0.9665*
Spain	Q21965–Q12020	79.31	3.59	74.58	83.23	−0.36	1.71	0.951*
Switzerland	Q21967–Q12020	83.52	3.21	79.64	87.70	−0.12	−0.41	0.9885***
United States	Q11967–Q42019	80.10	4.23	75.03	85.51	−0.18	−0.17	0.988***

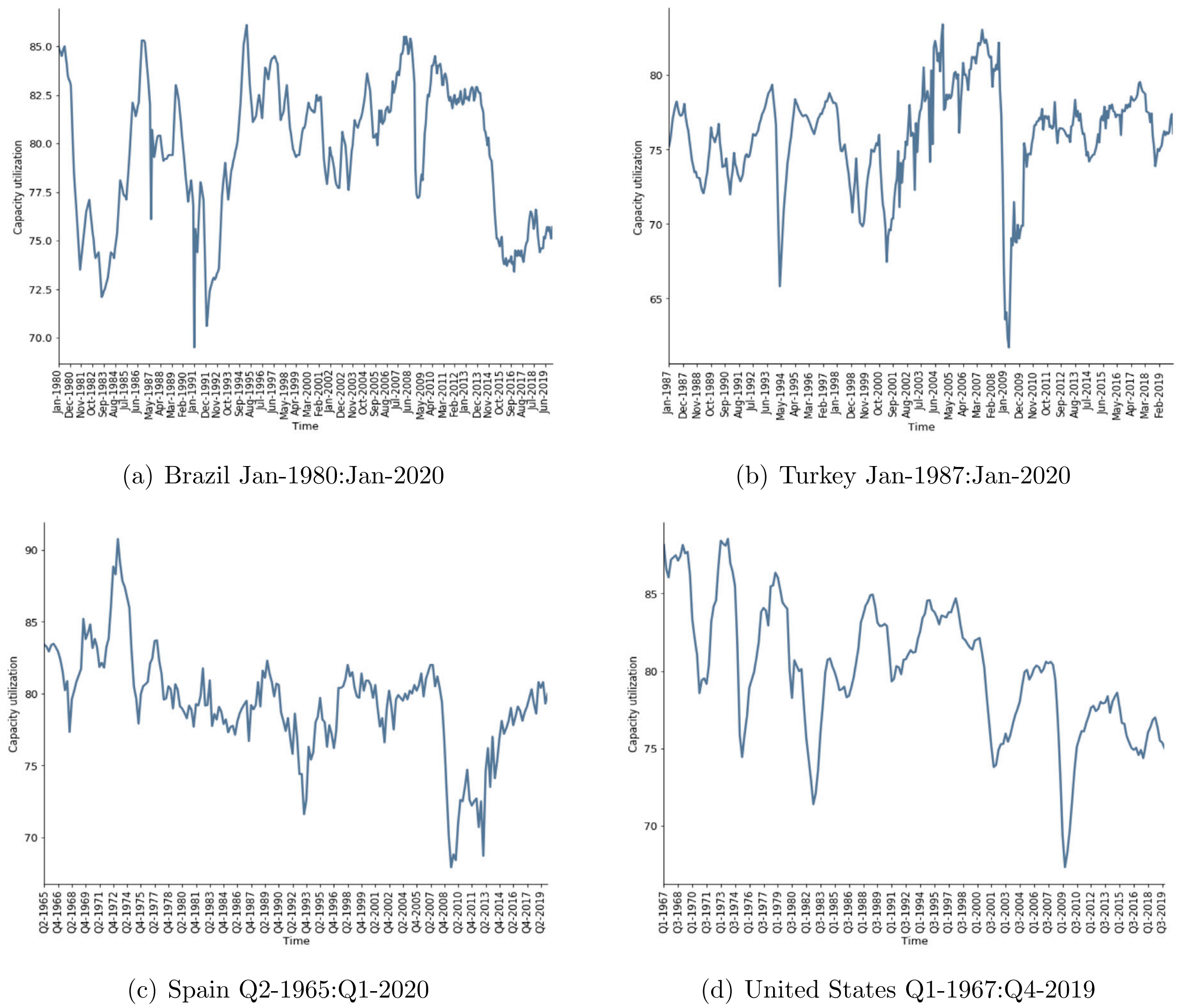
*Significant at: 1%.
**Significant at: 5%.
***Significant at: 10%.

Mexico, Peru, Philippines, Thailand, Turkey, and the United States. The quarterly sample is composed of France, Germany, Italy, New Zeeland, Spain, Switzerland, and the United States. The United States data is used in both samples as a benchmark for robustness checks. The quarterly data, and the montly data for Turkey was obtained in the Business Tendency Surveys database from OECD (2020); Argentina in INDEC (2020); Brazil in BCB (2020); Colombia in ANDI (2020); Mexico in BANXICO (2020); Peru in BCRP (2020); Phillipines in NSO (2020); Thayland in BOT (2020); and United States in FED (2020). Gahn and González (2022) presents a well-detailed description of how capacity utilization measures are estimated by each survey. For a critical discussion see Nikiforos (2016).

Table 1 details some descriptive statistics of the series. The time horizon in each country varies from seventeen years in the Philippines to fifty years in Germany. The standard deviation of the capacity utilization in Peru is relatively high, at 18%. In all the other countries the standard deviation is within expected ranges along with the mean relatively close to the median. Recessions are viewed as data realizations in the lower conditional deciles, while expansions are viewed as data realizations in upper conditional deciles. For most countries, the difference between the 1st to the 9th decile is at least two times the standard deviation. There are sufficient variations in all the series. In addition, most country data presents negative skewness indicating that most of the time the rate of capacity utilization stood below the average over the period. The kurtosis and the Shapiro–Wilk test for normality indicate that the data does not fit a normal distribution. All the time series were seasonally adjusted.

Fig. 2 presents the rate of capacity utilization in four selected countries. It is readily seen as a large negative effect of the Great Recession followed by a slow recovery. In Brazil, the political/economic crises of 2015 have considerably reduced the level of capacity utilization. The fall was relatively more intense than that of the Great Recession. After 2009, the Spanish rate of capacity utilization took more than ten years to achieve the prior recession levels. The rate of capacity utilization in the United States presents a significant negative trend (Gahn, 2020).

Possible structural breaks, the negative skewness, and the departure from normality presented in both samples highlight the benefits of using a QAR modeling-based approach, which is robust in the presence of most of these characteristics of the data (Koenker and Xiao, 2004). The application of a QAR model to the context of capacity utilization fluctuations is described in the next section.



(a) Brazil Jan-1980:Jan-2020

(b) Turkey Jan-1987:Jan-2020

(c) Spain Q2-1965:Q1-2020

(d) United States Q1-1967:Q4-2019

Fig. 2. Capacity utilization in Brazil, Turkey, Spain, and the United States.
Source: Multiple sources; See Section 3 p. 8.

3.1. Identification strategy

The literature dealing with the presence of unit root in the rate of capacity utilization focuses on constant-coefficient time series models (Gahn and González, 2022, 2020). The actual long-run value of utilization is treated as a good approximation of desired utilization. This makes sense if one assumes that the actual utilization cannot deviate persistently from the desired rate. The presence of unit root in the series indicates that shocks are persistent and that desired utilization may be endogenous to the actual utilization; If the series is stationary, shocks in utilization are transitory. Mean reversion indicates that u converges towards u^d. This approach is analogous to a conventional view of a branch of the business cycles literature which supposes that supply-side and structural components generate a steady trend in real GDP, and demand-side shocks fluctuations around the trend (Hamilton, 1989; Cerra and Saxena, 2017). For theoretical consistency and comparative purposes with related empirical literature, we interpret capacity utilization fluctuations around the long-run level as demand shocks.

The rate of capacity utilization is then modeled by an autoregressive, AR, process. An AR(q) process for the rate of capacity utilization is written as:

$$u_t = \omega + \sum_{i=1}^{q} \beta_i u_{t-i} + \epsilon_t, \tag{11}$$

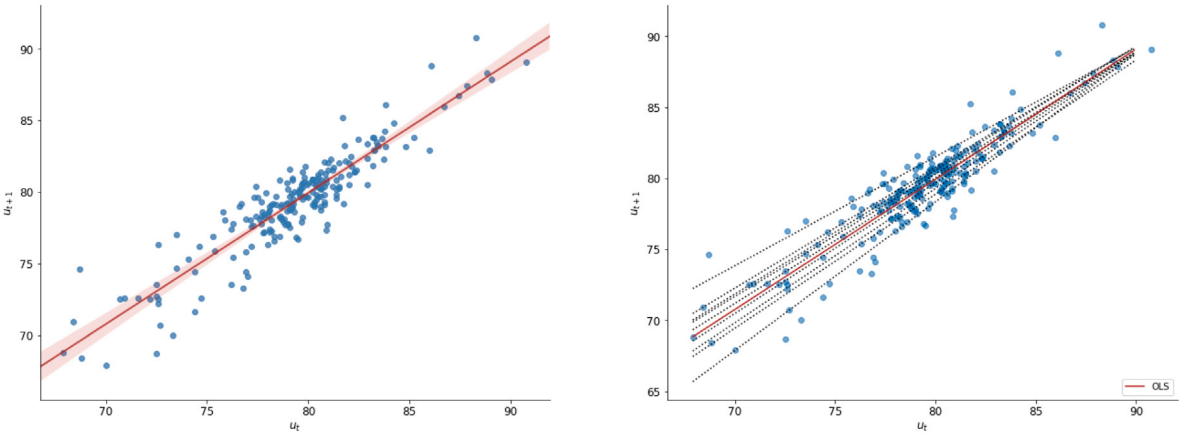
in which ω is a drift term, ϵ_t is the error term, and q is the lag order. The persistence in capacity utilization fluctuations is measured by $\sum_{i=1}^{q} \beta_i u_{t-i} = \alpha$. This parameter is the focus of the current empirical approaches. Conventionally, (11) can be rewritten as:

$$u_t = \omega + \alpha u_{t-1} + \sum_{i=1}^{q-1} \phi_i \Delta u_{t-i} + \epsilon_t. \tag{12}$$

Conventional methods can be used in (12) to test the unit root hypothesis. If $\alpha = 1$, then capacity utilization has a unit root; if $\alpha < 1$, capacity utilization is stationary. The usual procedure for estimation of (12) relies on a single measure of conditional central tendency, the mean. For illustrative purposes on the procedure, Fig. 3 panel (a) plots the AR(1) estimates for the rate of capacity utilization in Spain.

It is readily seen that the OLS regression line does not fit consistently with the scatter of points. The line regression at the median points is giving a good estimate of u_{t+1}, but the realizations in the lower conditional quantiles have a relatively higher variance. The utilization realizations that are high or low relative to utilization realizations in the previous periods are homogeneously treated along with the estimation procedure.

In Fig. 3 panel (b), the OLS regression line is compared with a quantile regression estimation (Koenker and Bassett, 1978). Fitting the data using this somewhat more inclusive model by introducing nine deciles minimizes the problem of heteroscedasticity. In addition, quantile regression is relatively more robust against outliers and when the



(a) Spain - OLS AR(1). (b) Spain - QAR(1).

Fig. 3. OLS AR(1) and QAR(1) of the rate of capacity utilization in Spain. Note: The red shading in panel (a) represents the confidence interval with 95% of significance.

data does not fit a normal distribution. The use of different measures of central tendency and dispersion can be useful to perform a more comprehensive analysis of how exogenous shocks can affect capacity utilization. This is the main idea of the QAR model which bases the identification strategy.

Hence, (12) is estimated by using a QAR procedure. The τ-th conditional quantile is defined as the value $Q_\tau(u_t|u_{t-1},\dots,u_{t-q})$ such that the probability that the utilization rate conditional on its lagged values will be less than $Q_\tau(u_t|u_{t-1},\dots,u_{t-q})$ is τ. If the capacity utilization value is low (high) relative to its lagged realizations, it means that a relatively large negative (positive) shock has occurred and that u_t is located below (above) the mean conditional on lag observations, $u_{t-1},\dots,u_{t-q}$, somewhere in the lower (upper) conditional quantiles (Koenker and Xiao, 2004). The QAR(q) process at the quantile τ of capacity utilization can be written as:

$$Q_\tau(u_t|u_{t-1},\dots,u_{t-q}) = \omega(\tau) + \alpha(\tau)u_{t-1} + \sum_{i=1}^{q-1}\phi_i(\tau)\Delta u_{t-i} + \epsilon_t. \quad (13)$$

Koenker and Xiao (2004) show that (13) can be estimated by using the Koenker and Bassett (1978) quantile regression estimator, which is based on the minimization of the quantile loss function. The estimation of (13) at different quantiles, $\tau \in (0,1)$, results in a set of estimates of the persistent measure, $\alpha(\tau)$. It is possible to test the null hypothesis of $\alpha = 1$ at different quantiles. This identification strategy also follows Galvao (2009), which extended the test to include deterministic components. The procedure is individually applied to all countries. The United States data is a special case. As Fig. 2 suggests, the data shows a small negative tendency. An expanded version of (12) with the trend cannot be rejected at the usual significance levels. The following particular version of (13) was estimated for the United States data:

$$Q_\tau(u_t|u_{t-1},\dots,u_{t-q}) = \omega(\tau) + \alpha(\tau)u_{t-1} + bt + \sum_{i=1}^{q-1}\phi_i(\tau)\Delta u_{t-i} + \epsilon_t. \quad (14)$$

Regarding inference, the null hypothesis is tested using the t-statistic for the quantile regression estimator, $\hat{\alpha}(\tau)$, expressed as:

$$t_n(\tau) = \frac{f(\widehat{F^{-1}(\tau)})}{\sqrt{\tau(1-\tau)}}(U_{-1}'M_Z U_{-1})^{1/2}(\hat{\alpha}(\tau)-1), \quad (15)$$

in which $f(.)$ and $F(.)$ are the probability and cumulative density functions of the error term, U_{-1} is the vector of lagged utilization rate, and M_Z is the projection matrix onto the space orthogonal to $Z = (1, t, \Delta u_{t-1},\dots,\Delta u_{t-q+1})$. The details of the estimation procedure to obtain $f(\widehat{F^{-1}(\tau)})$ can be checked in Koenker and Xiao (2004). Based on Koenker and Xiao (2004) and Galvao (2009), the critical values of $t_n(\tau)$ for different quantiles are simulated by a bootstrap of $R = 1000$ replications. The length of time of a shock in the utilization rate is estimated using the conventional (Andrews, 1993) procedure. The estimation is based on the half-life of a unit shock, $HL[\hat{\alpha}(\tau)] := log(1/2)/log[\hat{\alpha}(\tau)]$. This number characterizes the likely duration of an exogenous shock in the utilization rate in each specified quantile.

4. Results and discussion

The analysis begins with the Augmented Dickey–Fuller, ADF, and the KPSS unit root tests presented in Table 2, both to verify whether capacity utilization is a global stationary variable and as a benchmark by which the results from the QAR model are judged. The choice of both tests is for comparative purposes. In the ADF test, the null hypothesis of unit root is rejected for 13 countries. In Argentina, Peru, and the Philippines the null hypothesis cannot be rejected at the conventional levels of significance. Recall from Table 1 that Argentina and Peru presented relatively large short-run variations in the period. In general, ADF results closely replicate the evidence that capacity utilization is globally stationary (see, e.g., Gahn and González, 2022).

The results of the KPSS unit root test, in turn, suggest that the evidence of stationarity for some countries is unstable. The KPSS-statistics for Colombia, Mexico, New Zealand, and Spain contradict the ADF results. This conclusion is in line with Nikiforos (2020), which argues that some econometric results, as those reported in Gahn and González (2020), are not robust to more robust specifications of the unit root tests. In some sense, variable results should not surprise. The power of each unit root test varies according to the sample size, the presence of heteroscedasticity, and the non-normality of the series (see Table 1). These characteristics require the use of more general empirical approaches, such as the QAR model.

Now we reconsidered the series of capacity utilization now using the proposed QAR model. Tables 3–4 present the estimates of the largest autoregressive root for nine deciles. Given the relatively small number of observations, the lag order selection follows the results of a modified Akaike information criterion, MAIC (Ng and Perron, 2001). The rejection of the null hypothesis of unit root occurs when the t-statistic is less than or equal to the critical value, as usual. The significance level is 5%. Table 3 reports the results for the quarterly data, while Table 4 for the monthly data. The parameters are not strictly comparable among countries and samples.

To facilitate visualization, Fig. 4 plots the largest autoregressive root at each decile for the quarterly data. The blue points represent the transitory shocks (the rejection of the null hypothesis in Table 3), while

Table 2
ADF and KPSS unit root tests for the rate of capacity utilization.

Country	Data	q	ADF-Statistic	KPSS-Statistic
Argentina (ARG)	Monthly	10	−1.40	0.37***
Brazil (BRA)	Monthly	6	−3.18**	0.22
Colombia (COL)	Monthly	6	−3.74*	0.45***
Mexico (MEX)	Monthly	8	−2.94*	0.71**
Peru (PER)	Monthly	10	−0.53	1.86**
Philippines (PHL)	Monthly	10	−2.44	1.32**
Thailand (THA)	Monthly	8	−4.41*	0.12
Turkey (TUR)	Monthly	10	−4.42*	0.16
United States (USA)	Monthly	9	−3.48*	0.15
France (FRA)	Quarterly	4	−4.01*	0.09
Germany (DEU)	Quarterly	5	−5.99*	0.14
Italy (ITA)	Quarterly	4	−4.22*	0.08
New Zeeland (NZL)	Quarterly	4	−3.26**	0.58**
Spain (ESP)	Quarterly	4	−3.12**	0.68**
Switzerland (CHE)	Quarterly	5	−4.73*	0.17
United States (USA)	Quarterly	4	−3.84*	0.13

Note: Lags (q) are selected using the MAIC criterion.
The ADF tests the null hypothesis that unit root is present in capacity utilization.
The KPSS tests the null hypothesis that capacity utilization is stationary.
*Significant at: 1%.
**Significant at: 5%.
***Significant at: 10%.

the red points represent the persistent shocks (the non-rejection of the null hypothesis in Table 3). There are considerable variations among countries, except for shocks located around the median. The evidence on persistence is relatively clear. For most countries, we cannot reject the hypothesis that shocks below the median are persistent. In general, the shocks located from the median are transitory. The parameters associated with a transitory shock are relatively smaller than those associated with a persistent shock but with some exceptions. The positive shocks located from the 7th decile are persistent in Switzerland. In fact, the exclusion of Switzerland in panel (b) highlights the negative asymmetry in most countries of the quarterly sample, composed of developed countries.

The duration of a transitory shock, as approximated by the half-life, varies among countries. The average between-country duration of a relatively large expansionary shock, at the 9th decile, is of 3.5 quarters (Table 3). The expansionary shock dissipates more quickly than a median shock, in which the average between-country duration is of 7 quarters. This result is consistent with the view that the capacity utilization does not present persistent long-run variations (Skott, 2012). For most of these countries, however, this evidence is located at the median and upper deciles, not reflecting the entire conditional distribution of quarterly data on capacity utilization. The concentration of persistent shocks in capacity utilization located below the median is a remarkable pattern in Fig. 4.

Fig. 5 plots the largest autoregressive root at each decile for the monthly sample. Panel (a) plots the results for all countries, while panel (b) focuses on developing countries, dropping the U.S. data. The monthly sample reinforces the empirical evidence reported in Fig. 4. In general, persistent shocks in capacity utilization are relatively concentrated below the median. This evidence is independent of the exclusion of U.S. data. Note that the results for the United States used as a benchmark, are relatively similar in both samples (Tables 3–4). This regularity suggests that the results are invariant according to the data frequency.

The evidence on the asymmetric dynamic in capacity utilization fluctuations is not so general and remarkable as that presented in Fig. 4 panel (b). The presence of persistent shocks located at the upper deciles in Argentina, Brazil, and Mexico (Table 4) helps to explain the differences between the distribution of points in Figs. 4 and 5. In Brazil, the shocks are symmetrically distributed along the conditional quantiles. Below the median, only the 1st decile (the most recessionary shocks) is persistent. Above the median all shocks are persistent. In turn, shocks in Argentina and Mexico present a positive asymmetry. This pattern is evident in Argentina, in which the hypothesis of global stationarity is rejected (Table 2). In Mexico, few realizations above the median are sufficient to ensure global stationarity, but the shocks located from the 6th decile are persistent.

The average between-country duration of a transitory shock at the median in the monthly sample is of approximately 19.5 months. This estimate is relatively close and equivalent to those obtained with the quarterly data. In turn, the average between-country duration of a relatively large expansionary shock, at the 9th decile, is of approximately 6 months. The average between-country duration of a large expansionary shock is relatively greater in the quarterly sample data composed of developed countries than in the monthly sample data composed mostly of developing countries. In other words, in this sample of countries, transitory expansionary shocks dissipate relatively quickly in developing countries.

Figs. 4 and 5 show that the evidence of asymmetry varies among countries. However, there are some countries in which the evidence of asymmetry is more remarkable. Tables 3–4 suggest that at least in 12 countries the evidence of asymmetric dynamics (positive or negative) is non-negligible. As an illustration of this pattern detailed in Tables 3–4, Fig. 6 plots the largest AR root together with 95% confidence bands at each decile of the utilization rate for Colombia, France, Spain, and Thailand. For comparative purposes, the black line shows estimates at the conditional mean (OLS).

The four selected countries exhibit negative asymmetry, in which relatively large recessionary shocks (at the 1st decile) return AR roots right close to one, while relatively large expansionary shocks (at the 9th decile) are well below one. Estimated AR roots are monotonically decreasing as we move from lower to higher deciles in Colombia and Thailand. In France, in turn, the AR roots estimated by the QAR model are above the AR root estimated at the conditional mean until the 8th decile. The OLS estimation adjusts relatively well to capacity utilization fluctuations from the 3rd to the 8th decile in Spain. However, at the tails of the conditional distribution, the adjustment is remarkably asymmetric.

The fact that evidence on asymmetric dynamics and local persistence in capacity utilization fluctuations vary among countries should not surprise. Different mechanisms and eventually distinct effects for the same mechanisms over time helps to explain the lack of an unambiguous global pattern. However, some regularities emerge from the joint analysis of both samples (Tables 3–4). We find evidence that positive shocks located at the 6th and 7th deciles are transitory for 11 countries. Positive shocks located at the 8th and 9th deciles are transitory in 10 countries. The opposite regularity characterizes the negative shocks located below the median. Recessionary shocks located at the 1st and 2nd deciles are transitory only in 3 countries. Shocks located right below the median are transitory in 5 countries. Moreover, in 12 of 15 countries, there is evidence that the shocks located at the median are transitory.

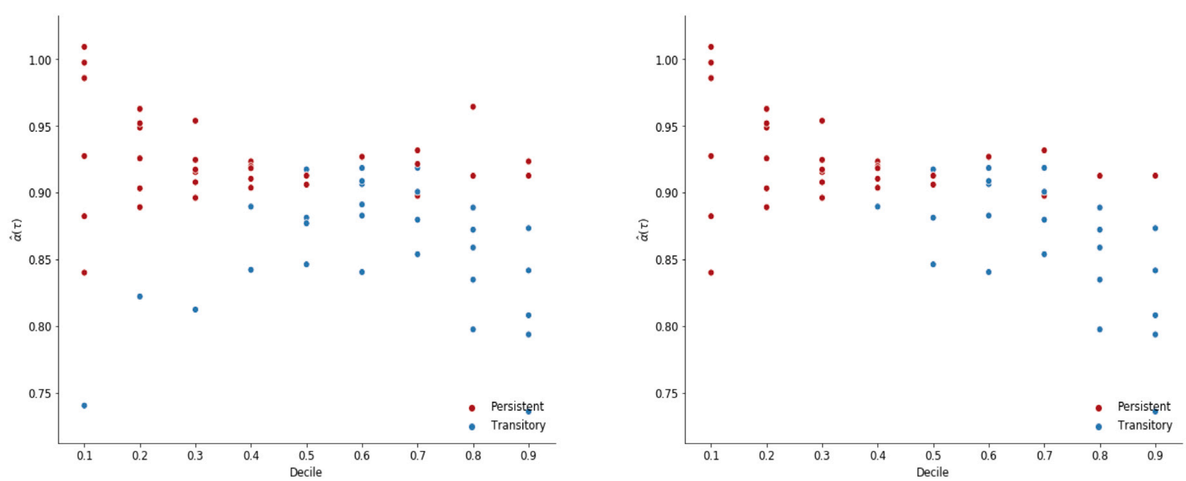
A transitory shock means that, after a disturbance, the actual utilization returns to its previous long-run level. Some studies analyzing the pattern of survey measures of capacity utilization, such as Gahn and González (2022), suggest that convergence towards the long-run level is a global characteristic of utilization fluctuations. In the present study, for most countries of the sample, the convergence towards the long-run previous values of capacity utilization are concentrated at the median and upper deciles. Hence, above the actual long-run level, an exogenous shock promotes a gap between actual and desired utilization that potentially induces firms to revise their rate of accumulation in order to increase available capacity. In this case, the actual utilization eventually adjusts towards the desired rate.

For most countries in the available sample, the new result is that if a shock occurs below the median, thus being negative or large recessionary (as in the 1st decile), the actual utilization does not return to its

Table 3
Estimates of the largest AR root at each decile of capacity utilization (quarterly).

Country	q	τ	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
		$\hat{\alpha}(\tau)$	0.986	0.889	0.896	0.890	0.913	0.927	0.880	0.835	0.736
FRA	4	t-stat.	−0.106	−1.754	−2.049	−2.807**	−1.924	−1.638	−2.44	−2.903**	−5.025**
		C.V.	−2.618	−2.571	−2.627	−2.603	−2.668	−2.675	−2.556	−2.375	−2.281
		HL($\hat{\alpha}$)	∞	∞	∞	5.919	∞	∞	∞	3.836	2.259
		$\hat{\alpha}(\tau)$	0.998	0.949	0.908	0.923	0.917	0.919	0.919	0.872	0.842
DEU	5	t-stat.	−0.031	−1.276	−2.519	−2.609	−2.939**	−2.959**	−2.811**	−3.932**	−4.172**
		C.V.	−2.383	−2.601	−2.853	−2.784	−2.747	−2.675	−2.552	−2.422	−2.214
		HL($\hat{\alpha}$)	∞	∞	∞	∞	8.040	8.153	8.164	5.069	4.019
		$\hat{\alpha}(\tau)$	0.840	0.926	0.925	0.904	0.846	0.840	0.854	0.797	0.794
ITA	4	t-stat.	−1.225	−1.062	−1.696	−2.458	−4.152**	−3.635**	−3.602**	−3.394**	−2.420**
		C.V.	−2.424	−2.587	−2.561	−2.542	−2.571	−2.535	−2.517	−2.492	−2.295
		HL($\hat{\alpha}$)	∞	∞	∞	∞	4.151	3.986	4.385	3.062	2.998
		$\hat{\alpha}(\tau)$	0.882	0.952	0.954	0.920	0.881	0.883	0.898	0.859	0.913
NZL	4	t-stat.	−1.393	−0.744	−0.837	−2.278	−3.512**	−3.553**	−2.591**	−3.019**	−1.224
		C.V.	−2.513	−2.559	−2.625	−2.511	−2.637	−2.608	−2.596	−2.574	−2.513
		HL($\hat{\alpha}$)	∞	∞	∞	∞	5.481	5.560	6.416	4.554	∞
		$\hat{\alpha}(\tau)$	1.009	0.963	0.915	0.910	0.913	0.906	0.901	0.913	0.808
ESP	4	t-stat.	0.139	−0.752	−1.957	−2.146	−2.426	−2.833**	−3.336**	−1.936	−2.543**
		C.V.	−2.419	−2.598	−2.572	−2.646	−2.649	−2.555	−2.548	−2.493	−2.367
		HL($\hat{\alpha}$)	∞	∞	∞	∞	∞	7.053	6.621	∞	3.251
		$\hat{\alpha}(\tau)$	0.740	0.822	0.812	0.842	0.877	0.891	0.922	0.964	0.923
CHE	5	t-stat.	−2.4407**	−2.3532**	−3.9291**	−3.812**	−2.805**	−2.779**	−2.204	−0.619	−0.629
		C.V.	−2.196	−2.226	−2.652	−2.732	−2.684	−2.690	−2.705	−2.535	−2.464
		HL($\hat{\alpha}$)	2.305	3.538	3.334	4.033	5.281	6.012	∞	∞	∞
		$\hat{\alpha}(\tau)$	0.927	0.903	0.917	0.918	0.906	0.909	0.932	0.889	0.873
USA	4	t-stat.	−1.430	−2.188	−2.272	−2.467	−3.262**	−3.095**	−2.760**	−3.159**	−2.895**
		C.V.	−2.378	−2.945	−3.280	−3.293	−3.140	−2.941	−2.732	−2.749	−2.754
		HL($\hat{\alpha}$)	∞	∞	∞	∞	7.022	7.240	9.798	5.880	5.116

Note: Lags (q) are selected using the MAIC criterion. Bootstrap: R = 1.000. HLs are calculated when $\hat{\alpha}(\tau)$ is smaller than unity; otherwise, HLs are set as infinity.
** t-statistic and critical values (C.V.) for the 5% significance level.



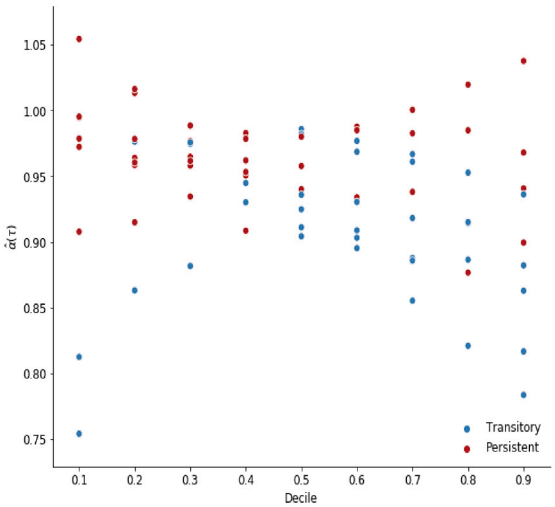
(a) All countries.

(b) All countries excluding Switzerland.

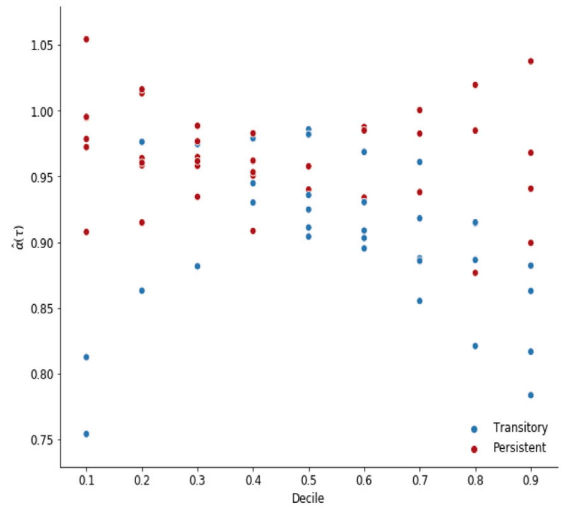
Fig. 4. Estimates of the largest AR root at each decile of capacity utilization (quarterly).

previous long-run level. This pattern is relatively remarkable in Fig. 4 panel (b), which shows the concentration of persistent shocks below the median. However, a negative shock need not be relatively high to produce a persistent change in capacity utilization fluctuations. In Brazil, for instance, during the worst recession of 2016, the fluctuation range of capacity utilization between a peak and a valley is near to 10%, while the new level has dropped to less than 10% (Fig. 2). This evidence supports the Amadeo–Lavoi–Dutt argument, in the sense that realizations of relatively low actual utilization make firms potentially revise their long-run level. A possible explanation is that in face of a negative shock firms may believe that changes in available capacity are relatively costly when compared to the net profit rate at the revised desired level.

In examining the persistence of recessionary shocks in the US economy, Nikiforos (2021) note that the average rate of capacity utilization for the period 1979–2007 is 2.7% below its average during 1948–1979. He also noted that over the last two recoveries, the level of utilization rate has not recovered to its post-1980 peaks. In a broad and speculative way, Nikiforos (2021) also relates this persistent decline to lower demand by noting that the decline in utilization rate is most notable for industries that face increases in international competition. Our results show that even by using survey measures of capacity utilization, the persistence of recessionary shocks is not confined to the US experience.

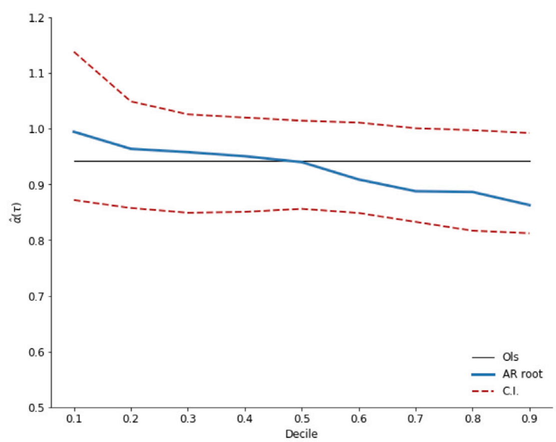


(a) All countries.

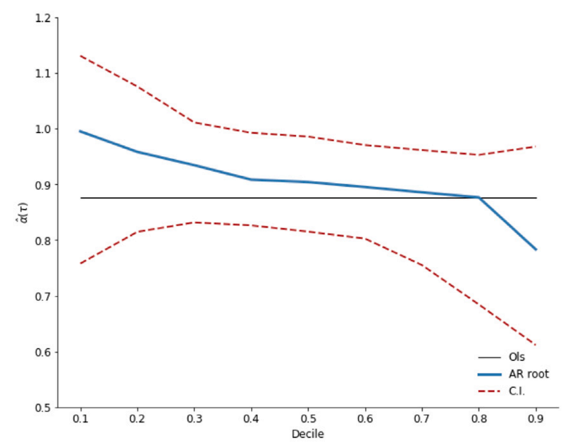


(b) Developing countries.

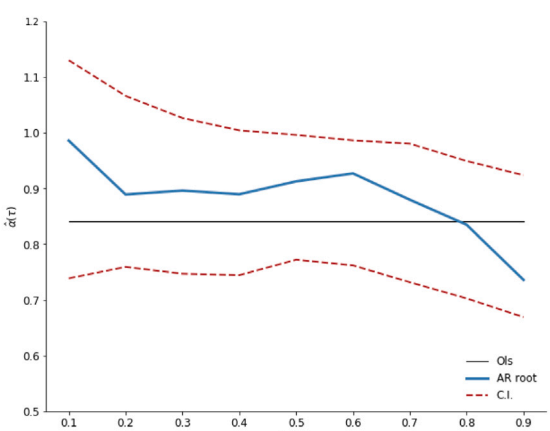
Fig. 5. Estimates of the largest AR root at each decile of capacity utilization (monthly).



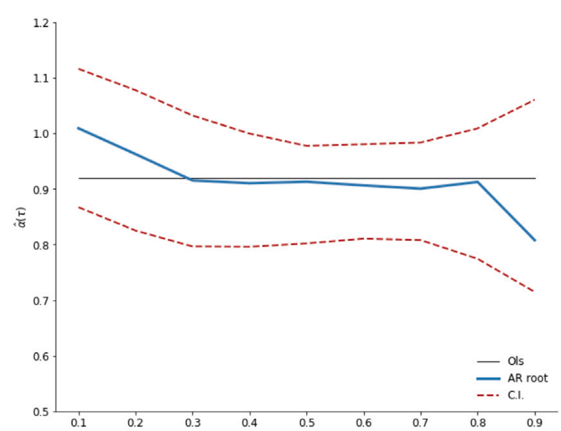
(a) Colombia.



(b) Thayland.



(c) France.



(d) Spain.

Fig. 6. AR(1) coefficients with 95% bootstrap confidence interval at each decile.

Table 4
Estimates of the largest AR root at each decile of capacity utilization (monthly).

Country	q	τ	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
		$\hat{\alpha}(\tau)$	0.754	0.915	0.965	0.962	0.958	0.988	1.000	1.020	1.038
ARG	10	t-stat.	−3.145**	−0.918	−0.52	−0.762	−1.163	−0.364	0.011	0.482	1.239
		C.V.	−2.711	−2.746	−2.599	−2.571	−2.541	−2.580	−2.453	−2.323	−2.324
		HL($\hat{\alpha}$)	2.455	∞	∞	∞	∞	∞	∞	∞	∞
		$\hat{\alpha}(\tau)$	0.979	0.976	0.975	0.979	0.986	0.985	0.983	0.985	0.968
BRA	6	t-stat.	−1.361	−2.6718**	−2.777**	−2.682**	−2.976**	−2.020	−1.964	−1.368	−1.740
		C.V.	−2.584	−2.595	−2.600	−2.668	−2.681	−2.586	−2.562	−2.626	−2.411
		HL($\hat{\alpha}$)	∞	28.899	26.941	32.659	48.466	∞	∞	∞	∞
	6	$\hat{\alpha}(\tau)$	0.995	0.964	0.958	0.951	0.940	0.909	0.888	0.886	0.863
COL		t-stat.	−0.068	−0.733	−1.065	−1.159	−1.482	−2.504**	−2.593**	−2.659**	−3.089**
		C.V.	−2.457	−2.647	−2.567	−2.633	−2.533	−2.487	−2.504	−2.395	−2.235
		HL($\hat{\alpha}$)	∞	∞	∞	∞	∞	7.248	5.819	5.748	4.697
		$\hat{\alpha}(\tau)$	0.812	0.863	0.882	0.930	0.925	0.934	0.938	0.915	0.941
MEX	8	t-stat.	−2.6012**	−3.1461**	−4.112**	−2.633**	−2.725**	−2.295	−1.888	−2.230	−0.690
		C.V.	−2.415	−2.467	−2.578	−2.499	−2.485	−2.540	−2.630	−2.576	−2.514
		HL($\hat{\alpha}$)	3.336	4.704	5.500	9.566	8.866	∞	∞	∞	∞
		$\hat{\alpha}(\tau)$	0.972	1.013	0.977	0.953	0.911	0.903	0.855	0.821	0.817
PER	10	t-stat.	−0.448	0.215	−0.393	−0.839	−3.180**	−3.285**	−2.977**	−2.926**	−2.976**
		C.V.	−2.775	−3.061	−3.077	−3.017	−3.080	−3.128	−2.952	−2.883	−2.927
		HL($\hat{\alpha}$)	∞	∞	∞	∞	7.445	6.801	4.431	3.512	3.421
		$\hat{\alpha}(\tau)$	1.054	1.016	0.989	0.983	0.982	0.969	0.961	0.914	0.882
PHL	10	t-stat.	1.359	0.818	−0.967	−1.729	−2.446**	−3.386**	−3.425**	−4.945**	−3.149**
		C.V.	−2.120	−2.133	−2.353	−2.335	−2.339	−2.444	−2.519	−2.563	−2.609
		HL($\hat{\alpha}$)	∞	∞	∞	∞	38.161	21.726	17.424	7.736	5.525
		$\hat{\alpha}(\tau)$	0.995	0.958	0.935	0.909	0.904	0.895	0.886	0.877	0.784
THA	8	t-stat.	−0.061	−0.658	−1.341	−2.257	−2.777**	−2.886**	−2.707**	−2.289**	−3.341**
		C.V.	−2.361	−2.487	−2.638	−2.556	−2.715	−2.649	−2.601	−2.493	−2.315
		HL($\hat{\alpha}$)	∞	∞	∞	∞	6.883	6.261	5.711	5.263	2.841
		$\hat{\alpha}(\tau)$	0.908	0.960	0.962	0.945	0.936	0.930	0.918	0.915	0.899
TUR	10	t-stat.	−1.211	−1.164	−1.639	−3.001**	−4.199**	−4.022**	−4.473**	−3.426**	−1.785
		C.V.	−2.568	−2.675	−2.721	−2.637	−2.534	−2.451	−2.417	−2.367	−2.317
		HL($\hat{\alpha}$)	∞	∞	∞	12.207	10.446	9.608	8.112	7.803	∞
		$\hat{\alpha}(\tau)$	0.979	0.978	0.976	0.978	0.980	0.977	0.967	0.953	0.936
USA	9	t-stat.	−1.424	−2.350	−3.131**	−2.969	−3.1192**	−3.281**	−4.035**	−4.996**	−5.784**
		C.V.	−2.757	−2.745	−2.818	−2.990	−3.044	−2.970	−2.990	−2.800	−2.666
		HL($\hat{\alpha}$)	∞	∞	28.177	∞	34.310	29.529	20.529	14.305	10.497

Note: Lags (q) are selected using the MAIC criterion.
Bootstrap: R = 1.000. HLs are calculated when $\hat{\alpha}(\tau)$ is smaller than unity; otherwise, HLs are set as infinity.
***t*-statistic and critical values (C.V.) for the 5% significance level.

The role of demand fluctuations in explaining the persistent decline in utilization rates in different countries is potentially more general.

In this sense, our new results are also consistent with a branch of empirical results of the business cycles literature. Some relevant contributions, such as those of Hamilton (1989) and Hosseinkouchack and Wolters (2013), find evidence that recessionary shocks have persistent effects on GDP fluctuations. According to Cerra and Saxena (2017), the lack of recovery is a widespread phenomenon across countries and periods. The significantly lower growth immediately after a recession holds for various income groups and regions. The negative asymmetry in capacity utilization fluctuations presented by most countries of our sample also finds support in this literature, which also shows that expansionary shocks are, in general, transitory. In a broad yet relevant sense, this qualitative difference between recessionary and expansionary shocks dates back at least to Keynes (1936). For Keynes, the length of time between the peak and a crisis or a recession, along with the business cycles, is much shorter than the length of time from the recession to the recovery.

This new evidence of negative asymmetry contrasts a theoretical contribution, usually evocated to justify the neo-Kaleckian long-run adjustment mechanism, that recognizes the possibility of a satisficing behavior from firms that will tolerate, within clear limits, deviation of the actual level from their desired level of utilization without modifying their behavior (Dutt, 1990). In an innovative contribution, Setterfield (2019) formalizes this argument by supposing that firms will change their rate of accumulation only when the utilization fluctuation exceeds lower or upper tolerable bounds. Out of the acceptable bounds, the current rate of capacity utilization adjusts towards the desired rate. In other words, relatively large negative or positive shocks are transitory. The notable concentration of persistent large recessionary shock (at the 1st decile) in our sample challenges this theoretical view.

The potential asymmetric coexistence of persistent and transitory shocks in capacity utilization fluctuations thus can be arguably related to hysteresis. But not necessarily in the usual neo-Kaleckian sense, in which the desired rate always varies directly with the actual rate of capacity utilization, ensuring that steady-state equilibrium conditions in the model are fully adjusted (Setterfield and David Avritzer, 2020). Hysteresis in the conventional sense that recessions may have persistent economic effects and welfare costs. In this case, persistent low aggregate demand fluctuations may make firms decide that low utilization becomes optimal. Under uncertainty, the optimal choice of firms is thus to under-utilize capacity.

5. Conclusions

This study empirically analyzes the presence of asymmetric dynamics and local persistence in capacity utilization fluctuations. To account for the differential impact of shocks of varying signs and magnitudes on capacity utilization, we employ a novel empirical strategy using a quantile autoregression (QAR) model. This approach constitutes the first contribution of this study to the empirical literature on demand-led growth and distribution, as capacity utilization fluctuations may exhibit asymmetry, with shocks below the median (associated with recessions or downturns) often having a more persistent effect compared to shocks above the median (associated with expansions or upturns).

The second contribution of this study is the robust empirical evidence supporting the existence of local unit root tendencies in capacity utilization, even when using official survey measures that are assumed to be globally stationary by construction. This evidence is consistent across all countries in the sample. The presence of persistent shocks in capacity utilization fluctuations, particularly in the lower quantiles associated with recessions, suggests that the cyclical adjustments of actual utilization are not simply temporary deviations from the long-run level. Instead, they indicate that these shocks themselves affect the long-run level of capacity utilization. The persistence of shocks in capacity utilization fluctuations, particularly in the lower quantiles associated with recessions, aligns with the predictions of the neo-Kaleckian long-run adjustment mechanism.

Our analysis also contributes to the utilization controversy by demonstrating the presence of asymmetric capacity utilization fluctuations in most selected countries. With a few exceptions in Brazil, Argentina, and Mexico, we find evidence of negative asymmetry, where shocks located below the median exhibit persistence, while those above the median are transitory. An important economic implication of asymmetric shocks in capacity utilization response is the distinct behavioral patterns during recessionary and expansionary periods. When capacity utilization is affected by persistent negative shocks, the economy may struggle to recover and reach previous desired production levels. Conversely, transient positive shocks may temporarily increase capacity utilization but may not necessarily lead to a revision in its long-term level. This finding aligns with the extensive empirical literature on business cycles and GDP fluctuations. However, it also suggests that the neo-Kaleckian long-run adjustment mechanism fails to explain positive variations above the median in capacity utilization. The theoretical generality predicted by its long-run adjustment mechanism lacks empirical support.

The results are limited by the small number of countries with available long-run time series of capacity utilization and by the quality of the data, especially those produced in developing countries. Therefore, caution should be exercised in generalizing the results and relating them back to theory. However, these findings raise intriguing questions that have significant implications for our understanding of the macroeconomy. The potential nonlinearity in the long-run adjustment mechanism of capacity utilization has important economic implications. If the response of capacity utilization to shocks is indeed asymmetric, with persistent negative shocks during recessionary periods and transitory positive shocks during expansionary periods, it suggests that the dynamics of capacity utilization are not simply movements along a fixed long-run level. From a policy perspective, the presence of asymmetric capacity utilization fluctuations implies that the management of recessions and expansions requires a differentiated approach. Persistent negative shocks that lead to prolonged periods of low capacity utilization may require targeted policies to stimulate demand and facilitate the recovery of the economy. On the other hand, transient positive shocks that temporarily increase capacity utilization may not necessarily warrant significant policy interventions, as they may not lead to a revision in the long-term level of capacity utilization.

Further research is warranted to enhance our comprehension of the underlying theoretical mechanisms that potentially elucidate the asymmetric dynamics of capacity utilization fluctuations and their macroeconomic ramifications. Additionally, the discovery that the neo-Kaleckian long-run adjustment mechanism falls short in elucidating positive variations exceeding the median in capacity utilization poses a challenge to the theoretical universality suggested by this adjustment. Consequently, it underscores the necessity for additional theoretical advancements aimed at integrating the observed nonlinearity in capacity utilization dynamics.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Guilherme de Oliveira reports financial support was provided by Fundação de Amparo a Pesquisa e Inovação do Estado de Santa Catarina.

Data availability

Data will be made available on request.

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